

HCAHPS Patient Analysis Project

Dataset Understanding Report

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1. Introduction

This report explains the HCAHPS dataset used in the project. It describes each table, its purpose, important columns, and how the tables are connected. Understanding the dataset helps in creating a correct data model and an accurate Power BI dashboard.

2. Dataset Overview

The HCAHPS dataset contains information about patient satisfaction collected from hospitals. It includes survey questions, quality measures, state information, reporting periods, response categories, national results, and state-level results. These tables work together to provide a complete view of patient experience.

3. Dataset Tables

Table Name	Description
Measures	Stores hospital quality measures used to evaluate patient experience and satisfaction.
Questions	Stores HCAHPS survey questions along with their corresponding top-box, middle-box, and bottom-box answer categories.
Reports	Stores reporting period information, including the release period and corresponding start and end dates.
Responses	Stores hospital survey response information, including completed surveys, response rates, facility identifiers, reporting periods, and states.
States	Stores U.S. state names and their corresponding geographic regions.
National Results	Stores national-level HCAHPS survey performance percentages for each quality measure and reporting period.
State Results	Stores state-level HCAHPS survey performance percentages for each quality measure and reporting period.

4. Reports Table

4.1 Purpose

The Reports table stores the reporting periods used in the HCAHPS survey. It contains the release period along with the survey start date and end date.

4.2 Business Description

The Reports table defines the time period for each HCAHPS survey report. It helps organize survey data by reporting period and allows comparison of patient satisfaction results across different years.

4.3 Primary Key

Release Period

The Release Period is the unique identifier for each HCAHPS report. It uniquely identifies every reporting period in the Reports table.

4.4 Relationships

- Related to the **Responses** table through the **Release Period** column.
- Related to the **National Results** table through the **Release Period** column.
- Related to the **State Results** table through the **Release Period** column.

4.5 Important Columns

Column Name	Description
Release Period	Unique identifier for each reporting period.
Start Date	Start date of the patient survey period.
End Date	End date of the patient survey period.

5. States Table

5.1 Purpose

The States table stores information about U.S. states and their regions. It helps identify the location of hospitals and supports state-wise and region-wise analysis.

5.2 Business Description

The States table provides the state code, state name, and region for each state. It is used to group and compare patient satisfaction results across different states and regions.

5.3 Primary Key

State

The State column is the unique identifier for each state. It is used to identify each state in the dataset.

5.4 Relationships

- Related to the **Responses** table through the **State** column.
- Related to the **State Results** table through the **State** column.

5.5 Important Columns

Column Name	Description
State	Two-letter code for each U.S. state.
State Name	Full name of the state.
Region	Geographic region where the state belongs.

6. Measures Table

6.1 Purpose

The Measures table stores information about the healthcare quality measures used in the HCAHPS survey.

6.2 Business Description

This table defines the quality measures used to evaluate patient satisfaction. Each measure has a unique ID and a measure type, which helps organize survey questions and analyze performance.

6.3 Primary Key

Measure ID

The Measure ID uniquely identifies each healthcare quality measure.

6.4 Relationships

- Related to the **Questions** table through the **Measure ID** column.
- Related to the **National Results** table through the **Measure ID** column.
- Related to the **State Results** table through the **Measure ID** column.

6.5 Important Columns

Column Name	Description
Measure ID	Unique identifier for each measure.
Measure	Name of the healthcare quality measure.
Type	Category of the measure.

7. Questions Table

7.1 Purpose

The Questions table stores all HCAHPS survey questions and links each question to a healthcare quality measure.

7.2 Business Description

This table contains the survey questions presented to patients. Each question belongs to a specific measure and includes the response categories used for analysis.

7.3 Primary Key

Question Num

The Question Num uniquely identifies each survey question.

7.4 Relationships

- Related to the **Measures** table through the **Measure ID** column.

7.5 Important Columns

Column Name	Description
Question Num	Unique identifier for each question.
Measure ID	Links the question to a measure
Question	Survey question text.
Bottom-box Answer	Lowest satisfaction response.
Middle-box Answer	Neutral satisfaction response.
Top-box Answer	Highest satisfaction response.

8. Responses Table

8.1 Purpose

The Responses table stores survey participation information for hospitals.

8.2 Business Description

This table records the number of completed surveys and the response rate for each hospital during a reporting period.

8.3 Primary Key

No single Primary Key

This table does not have a single-column primary key. A combination of **Release Period**, **State**, and **Facility ID** uniquely identifies each record.

8.4 Relationships

- Related to the **Reports** table through the **Release Period** column.
- Related to the **States** table through the **State** column.

8.5 Important Columns

Column Name	Description
Release Period	Survey reporting period.
State	State code.
Facility ID	Unique hospital identifier.
Completed Surveys	Number of completed surveys.
Response Rate (%)	Survey response rate.

9. National Results Table

9.1 Purpose

The National Results table stores patient satisfaction results at the national level.

9.2 Business Description

This table contains national survey percentages for each healthcare quality measure during every reporting period.

9.3 Primary Key

Composite Primary Key

The combination of **Release Period** and **Measure ID** uniquely identifies each record.

9.4 Relationships

- Related to the **Reports** table through the **Release Period** column.
- Related to the **Measures** table through the **Measure ID** column.

9.5 Important Columns

Column Name	Description
Release Period	Survey reporting period.
Measure ID	Healthcare quality measure.
Bottom-box Percentage	Lowest satisfaction percentage.
Middle-box Percentage	Neutral satisfaction percentage.
Top-box Percentage	Highest satisfaction percentage.

10. State Results Table

10.1 Purpose

The State Results table stores patient satisfaction results for each state.

10.2 Business Description

This table contains state-level survey percentages for each healthcare quality measure and reporting period. It helps compare healthcare performance across states.

10.3 Primary Key

Composite Primary Key

The combination of **Release Period**, **State**, and **Measure ID** uniquely identifies each record.

10.4 Relationships

- Related to the **Reports** table through the **Release Period** column.
- Related to the **Measures** table through the **Measure ID** column.
- Related to the **States** table through the **State** column.

10.5 Important Columns

Column Name	Description
Release Period	Survey reporting period.
State	State code.
Measure ID	Healthcare quality measure
Bottom-box Percentage	Lowest satisfaction percentage.
Middle-box Percentage	Neutral satisfaction percentage.
Top-box Percentage	Highest satisfaction percentage.

11. Fact Tables

Fact tables store measurable business data that is used for analysis. They contain numeric values and connect to multiple dimension tables through keys.

Fact Tables in this project

Fact Table	Reason
National Results	Stores national-level patient satisfaction percentages
State Results	Stores state-level patient satisfaction percentages.
Responses	Stores hospital survey response counts and response rates.

12. Dimension Tables

Dimension tables provide descriptive information that helps categorize and filter the data stored in the fact tables.

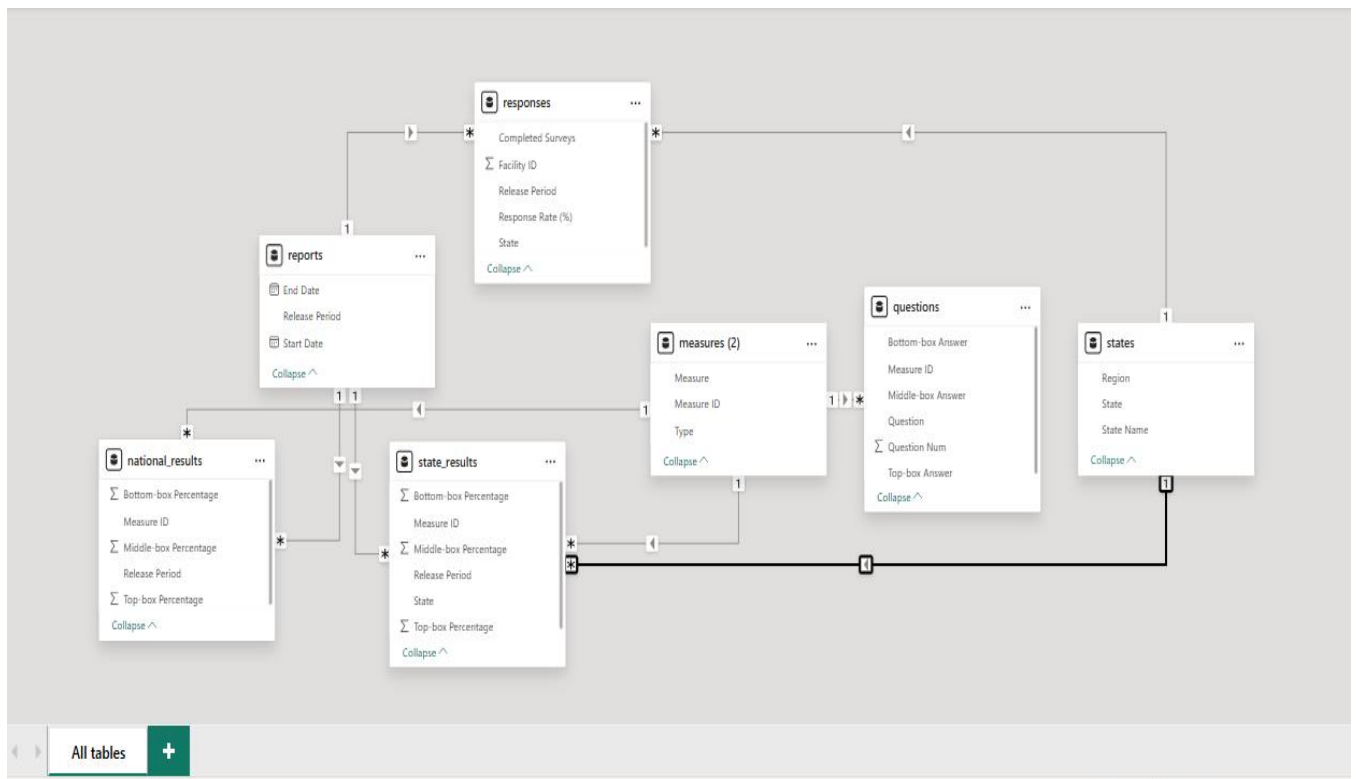
Dimension Tables

Dimension Table	Purpose
Reports	Stores reporting periods.
States	Stores state names and regions.
Measures	Stores healthcare quality measures.
Questions	Stores survey questions.

13. Star Schema Design

The HCAHPS dataset follows a star schema design. The fact tables contain measurable patient satisfaction data, while the dimension tables provide descriptive information such as reporting periods, states, measures, and survey questions. This design improves query performance and makes dashboard development in Power BI easier.

14: Relationship Diagram



Description

The Power BI data model was designed using a star schema approach. The Reports, Measures, and States tables act as dimension tables, while the Responses, National Results, and State Results tables contain analytical data. The relationships between these tables enable efficient filtering, accurate KPI calculations, and interactive dashboard reporting.